

Observable Classification Theory for Sectorized Observable Frameworks

Wall Records, Wall Coordinates, Taxonomy, and Smooth Local Models

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This paper is Part XI of the RIME program. Paper VII isolates generic accessibility completion; Paper VIII fixes the static SOF object layer; Paper IX studies observable dynamics over SOFs; and Paper X isolates the Universal Observable Pipeline together with a five-layer SOF Registry. Paper XI turns that registry into an observable classification layer: given registered SOFs, what feature space classifies their observable behavior?

Abstract

Problem. The SOF Registry records species, SOF objects, observable ladders, dynamics, and diagnostics. This makes cross-species comparison possible, but it does not yet say what type a registered SOF has at the observable level. The next question is not whether every wall belongs to a known singularity class. The question is which observable signatures and diagnostics organize the behavior of registered SOFs.

Approach. We introduce a five-layer organization. First, a **per-observable wall record** preserves the observation history of a chosen observable along a declared deformation: its change locus, discriminant data, diagnostics, and ordered atomic events $W_{a,1}, \dots, W_{a,k_a}$. The **aggregate wall record** collects these histories across the declared ladder. Second, feature extraction derives a **wall signature** from each atomic event: rank type, support type, repair type, oscillation type, plateau type, or equivalently changes such as $(\Delta R_1, \Delta R_2, \Delta D, \Delta \tau, \Delta P_d)$. The multiset of all atomic-event signatures is the **wall spectrum** $\text{Spec}_W(\mathcal{F}, \mathcal{L}, \Gamma)$. Third, a **wall-coordinate profile** records position, observable orientation, regularity, and stochastic status. Fourth, the **Observable Wall Taxonomy** groups wall spectra into collision, repair, terminal-structure, plateau/rate, nonsmooth/discrete, and bridge/incidence wall families. Smooth ADE models enter only afterward as the fifth layer: candidate local normal forms for sufficiently smooth discriminant branches.

Results. The main result is a proto-geometric wall-record schema, not a complete classification theorem. It separates mechanism labels from geometric coordinates and turns wall records into computable features such as crossing count, atomic-event count, wall-cell count, repair index, oscillation index, depth index, rate ratio, spectral gap, and codimension. Four extended profiles expose four different geometries: a constructed real-symmetric endpoint gives two isolated order-one A_1 -type pair-gap closures; nested Erdos–Renyi percolation gives a monotone opening window near $p = 0.08\text{--}0.10$; matched Kuramoto ensembles give a freezing crossover near $K = 1.6\text{--}1.8$; and continuous weakening of one GRN repression edge gives a two-to-one terminal-basin loss in the controlled bracket $[0.520, 0.525]$. A matched fixed-time CLE/SSA control withdraws the earlier GRN noise-wall interpretation. The extended census retains the original 24 records, adds these four records, and contains 28 records, 38 class memberships, and 19 first-pass eligible records. All six classes meet the predeclared three-record, two-species, two-deformation finite-sample target. Class A closes that target through a constructed non-Rubik witness; naturally occurring non-Rubik Class A breadth remains open.

Implications. Paper XI develops the zeroth geometric layer of SOF wall theory. The wall-record language is source-independent, while realized wall geometry is source-dependent. The

resulting wall-coordinate space has position, orientation, regularity, and stochasticity, but it does not yet have a metric, curvature, coordinate-change law, or transport structure. ADE-type local models may be useful for spectral collision or smooth discriminant maps, but they cannot be the default language for all SOF walls.

Revision note. Relative to the initial release, this version adds four controlled wall profiles, extends the original 24-record census to an independently generated 28-record census, and withdraws the earlier low- Ω GRN noise-wall interpretation after a matched CLE/SSA control. The A–F taxonomy is retained, while the distinction between observation histories and derived wall signatures is made explicit.

Notation Table

Symbol	Meaning
$\mathcal{F} = (V, \{Q_i\}, \mathcal{X})$	Sectorized Observable Framework
$\mathcal{F}_t$	SOF deformation
$O(t)$	observable trajectory or shadow
Σ_O	wall or discriminant locus of O
Σ_{comm}	commutativity locus in generator-weight moduli
Δ_O	target discriminant in observable space
$\mathcal{W}_a(\mathcal{F}, \mathcal{L}, \Gamma)$	per-observable wall record for $O_a \in \mathcal{L}$
$W_{a,j}$	the j th atomic wall event in the record of O_a
$\mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma)$	aggregate wall record $(\mathcal{W}_a)_{O_a \in \mathcal{L}}$
Feat_W	feature-extraction map from wall records or atomic events to derived wall features
$\text{Sig}_W(W_{a,j})$	computable signature of an atomic wall event
$\text{Spec}_W(\mathcal{F}, \mathcal{L}, \Gamma)$	multiset of atomic-event wall signatures
$\text{Prof}_W(\mathcal{W})$	wall profile: class together with position, orientation, regularity, and stochasticity
Pos, Ori, Reg, Sto	wall-coordinate components
$\text{Term}(\mathcal{F}_t)$	terminal decomposition induced by the declared dynamics and observable realization
WallAssign	record-producing assignment (bookkeeping): $(\mathcal{F}, \mathcal{L}, \Gamma) \mapsto \mathcal{W}$
$N_{\text{cross}}, N_{\text{evt}}, N_{\text{cell}}$	crossing-step count, atomic-event count, and wall-cell count
R_W, p_W, O_W, P_W	repair, persistence, oscillation, and plateau indices
$\bar{D}, \rho_W, \gamma_W, c_W, \delta_W$	depth, rate-ratio, spectral-gap, codimension, and density diagnostics
R_1, R_2, D	accessibility observable ladder
$\tau(O)$	characteristic time scale when defined
$P_d(t)$	plateau function
A_k, D_k, E_k	Arnold ADE singularity families

1 Introduction

The post-Paper-VII SOF arc has three layers:

Paper	Role
Paper VIII	SOF object layer
Paper IX	observable trajectories over SOFs
Paper X	Universal Observable Pipeline and five-layer SOF Registry

Paper XI asks what comes after the registry.

The registry is a taxonomy input:

species $\rightarrow$ SOF object $\rightarrow$ observable ladder $\rightarrow$ dynamics $\rightarrow$ diagnostics.

Paper X records which SOFs exist and how they enter the observable pipeline. Paper XI asks what observable type those SOFs have. In other words, the registry is a list of realized species, while Paper XI extracts the observable features by which those species can be compared.

Paper XI asks whether registered wall records can be grouped into classes and compared geometrically. The answer is not yet a complete classification. The safe claim is that the classification problem splits into observable signatures, wall spectra, wall coordinates, mechanism classes, normal forms, and smooth local models.

This paper is deliberately conservative. It does not claim that Arnold ADE classification applies to all SOF walls. It treats ADE as one candidate local model for sufficiently smooth discriminant-like wall records and records where that language does not apply. The organizing object is the wall record, not an ADE class.

The RIME papers up to this point mostly control **wall location**: where support, rank, accessibility, plateau, or collision behavior changes. They do not yet give a general wall-crossing law for how all invariants transform across every wall. Paper XI therefore asks a prior organizational question: what kinds of observable walls have been registered, and how should they be organized before any transformation law is attempted?

The answer has five layers:

Layer	Object	Question
Wall Records	loci and observable changes	Where do walls occur?
Wall Signatures	computable diagnostics of events and records	What does the wall look like?
Wall Coordinates	position, orientation, regularity, stochasticity	How is the wall realized?
Observable Wall Taxonomy	classes of signatures	What type of wall is it?
Smooth Wall Models	local smooth-discriminant models	When can ADE-like tools apply?

For later diagnostic reports, the same layer becomes a feature extractor. An application-facing SOF report need not use theorem proofs directly. It can emit quantities such as crossing count, wall-cell count, repair index, plateau or oscillation index, and depth index, then interpret them as observable diagnostics.

This also extends the categorical architecture begun in Paper VIII:

Source $\rightarrow$ SOF_{str} $\rightarrow$ Observable ladder
 $\rightarrow$ Observable dynamics $\rightarrow$ Wall records
 $\rightarrow$ Feature extraction $\rightarrow$ Wall signatures and diagnostics
 $\rightarrow$ Wall coordinates $\rightarrow$ Wall taxonomy.

The program invariant of this paper is therefore:

Wall-record language is source-independent; realized wall geometry is source-dependent.

The coordinate language does not assert that wall records already form a manifold or phase space. It supplies a comparison chart before any metric, curvature, or transport law has been defined.

Paper XI does not prove that this last passage is a categorical invariant of SOFs. It introduces the record-producing layer needed to make the passage explicit once the observable ladder and deformation geometry have been chosen.

2 Boundary Notes

Paper XI is organized by constructive definitions, record audits, and explicit non-claims rather than by a single classification theorem. Its scope is deliberately record-level. In particular, it does not claim:

1. a classification of the underlying dynamical systems, phase transitions, singularities, or bifurcations that produce the records;
2. a universal wall-crossing or wall-transition law;
3. that the qualitative wall-coordinate space already carries a metric, topology, curvature, or coordinate-change structure;
4. that Arnold ADE models apply outside sufficiently smooth discriminant-like branches;
5. that WallAssign is invariant under all weak SOF comparisons, path reparametrizations, thresholds, or sampling refinements.

These boundaries are repeated locally where a construction or computational profile requires a narrower claim. The repeated boundary statements are part of the taxonomy: they record which conclusions each wall signature licenses and which native interpretations remain external to SOF.

3 Wall Records

3.1 Definition

For a declared ladder observable $O_a \in \mathcal{L}$, its **per-observable wall record** is the data

$$\mathcal{W}_a(\mathcal{F}, \mathcal{L}, \Gamma) = (\mathcal{F}_t, O_a(t), \Sigma_{O_a}, \Delta_{O_a}, \mathcal{D}_{O_a}),$$

where:

1. $\mathcal{F}_t$ is a specified SOF deformation;
2. $O_a(t)$ is an observable shadow or continuous observable field;
3. Σ_{O_a} is the locus where the observable changes qualitative type;
4. Δ_{O_a} is the target discriminant when a pullback description exists;

5. $\mathcal{D}_{O_a}$ is the raw measured diagnostic history, such as $\tau(O_a)$, repair, plateau, frozen-pair count, or spectral-gap change.

A wall record is an **observation history** relative to the declared SOF realization, observable ladder, deformation path, and audit convention. It retains what was observed, where it changed, and in what path order. It is not itself a wall signature: signatures arise only after a feature-extraction rule has been chosen.

If the chosen path order on Γ meets k_a registered event components, the associated event sequence is

$$\text{Evt}(\mathcal{W}_a; \Gamma) = (W_{a,1}, W_{a,2}, \dots, W_{a,k_a}).$$

The $W_{a,j}$ are atomic wall events. Paper XI classifies their signatures and the aggregate records built from them; it does not identify the event, the per-observable record, and the aggregate record as the same object.

3.2 Wall Assignment

Paper VIII defines strict SOF morphisms and the category SOF_{str} . Paper XI adds a record-producing construction

$$\text{WallAssign} : (\mathcal{F}, \mathcal{L}, \Gamma) \mapsto \mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma),$$

read as: a realized SOF $\mathcal{F}$, together with a chosen observable ladder $\mathcal{L}$ and deformation geometry or sampled path Γ , determines the aggregate family of per-observable wall records along that geometry.

This notation is intentionally modest. It is a bookkeeping assignment, not a claim that wall records form a categorical morphism out of SOF_{str} . The present paper does not prove invariance under every weak SOF comparison. It only asserts that wall records are naturally recorded after SOF realization, observable extraction, and deformation choice.

3.3 Aggregate Wall Record

For a realized SOF deformation, the aggregate wall record is the indexed family

$$\mathcal{W}(\mathcal{F}, \mathcal{L}, \Gamma) = (\mathcal{W}_a(\mathcal{F}, \mathcal{L}, \Gamma))_{O_a \in \mathcal{L}}.$$

Schematically, its registered change loci may include

$$\{\Sigma_{O_a} : O_a \in \mathcal{L}\} = \{\text{rank jumps, support jumps, accessibility jumps, plateau intervals, collision loci, ...}\}.$$

For a one-parameter deformation, each per-observable record carries the ordered atomic-event sequence

$$\text{Evt}(\mathcal{W}_a; \Gamma) = (W_{a,1}, W_{a,2}, \dots, W_{a,k_a}),$$

where each $W_{a,j}$ is an atomic observable event encountered along the path. This sequence is analogous in spirit to a critical sequence in Morse theory, a bifurcation diagram, a persistence barcode, or an RG-flow history, but it is not identified with any of these objects. It is the SOF-native record of observable changes along a chosen deformation.

The ellipsis is important: the registry is open. A new wall type may enter only after its SOF object, observable ladder, deformation geometry, jump or discriminant locus, and diagnostic package are explicit.

Observable Wall Record Pipeline

Paper XI classifies registered observable change records, not native systems directly

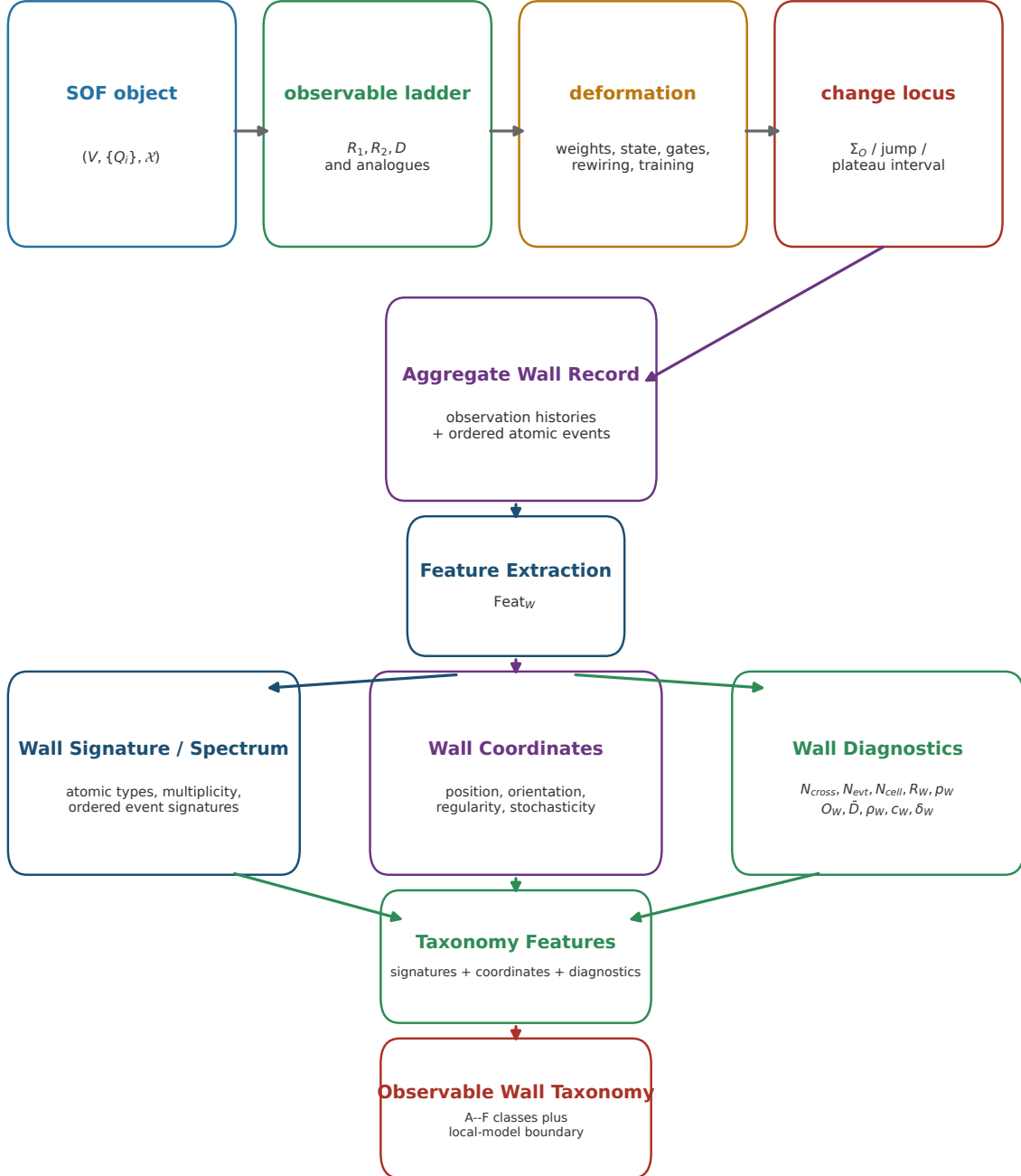


Figure 1. Observable wall-record pipeline. A realized SOF, observable ladder, deformation geometry, and change loci produce indexed per-observable histories and their aggregate wall record. Feature extraction derives atomic signatures, wall spectra, wall coordinates, and diagnostics; these derived taxonomy features, rather than the observation history itself, feed the Observable Wall Taxonomy. ADE is only a candidate local model on sufficiently smooth discriminant branches.

4 Wall Signatures

Wall records preserve what was observed along a deformation. A **wall signature** is a feature extraction from that history: it records what an atomic wall event looks like in computable terms. Schematically,

$$\mathcal{W} \xrightarrow{\text{Feat}_W} (\text{Sig}_W, \text{Spec}_W, \text{Prof}_W, \mathcal{D}_W).$$

Wall signatures are therefore derived from wall records and are not primitive SOF objects. Different admissible feature maps may extract different signatures from the same observation history, so the extraction convention belongs to the declared audit semantics.

The basic object is the signature of an atomic observable event

$$\text{Sig}_W(W_{a,j}) = (\text{rank type}, \text{support type}, \text{repair type}, \text{oscillation type}, \text{plateau type}).$$

For accessibility and trajectory diagnostics, the same information can be recorded as an observable-difference vector

$$\text{Sig}_W(W_{a,j}) = (\Delta R_1, \Delta R_2, \Delta D, \Delta \tau, \Delta P_d).$$

The first notation records qualitative type; the second records measured observable change. Paper XI classifies these observable signatures, not bare wall events.

A typical signature has the form

$$\text{Sig}_W(W_{a,j}) = (c, k, p, m, r, b, \dots),$$

where:

1. c records the number of detected components or cells;
2. k records codimension when a meaningful ambient geometry is known;
3. p records persistence: persistent, transient, terminal, or degenerate;
4. m records temporal profile: monotone, oscillatory, plateau, flat, or discontinuous;
5. r records repair status: repairable, unrepaired, terminal, or not applicable;
6. b records boundary type: collision, rank/support jump, absorbing boundary, activation kink, stopping boundary, or filtration degeneration.

This is not a complete invariant. It is a computable signature schema for comparing wall records without pretending that all walls belong to one smooth classification theory.

4.1 Wall Spectrum

The **wall spectrum** of a realized SOF deformation is the multiset of all atomic-event signatures observed across the chosen observable ladder:

$$\text{Spec}_W(\mathcal{F}, \mathcal{L}, \Gamma) = \{\{\text{Sig}_W(W_{a,j}) : O_a \in \mathcal{L}, 1 \leq j \leq k_a\}\},$$

where k_a is the number of registered events in the record of O_a . Equivalently, one may write schematically

$$\text{Spec}_W(\mathcal{F}, \mathcal{L}, \Gamma) = \{\{\text{rank wall}, \text{accessibility wall}, \text{plateau wall}, \dots\}\}.$$

The multiset notation allows repeated wall types. A species may exhibit two distinct repair walls, or both a collision wall and a plateau wall, and these should not be collapsed into a single label.

4.2 Observable Wall Diagnostics

Feature extraction also summarizes the measured history into numerical wall diagnostics. Their values are relative to the declared realization, ladder, deformation path, threshold, depth cutoff, and sampling protocol. Paper XI does not prove invariance under changes of those choices. Within a fixed audit convention, they provide reproducible coordinates for comparing wall spectra and for passing summaries to later SOF reports.

Diagnostic	Definition	Example audited value
Crossing count N_{cross}	number of sampled parameter transitions containing at least one atomic wall event	GridWorld obstacle path: 2 crossing steps; SIR and graph controls: 1 each
Atomic-event count N_{evt}	total number of atomic observable-status changes, including simultaneous changes and repeated changes of one pair	GridWorld obstacle path: 190 pair-status events
Wall-cell count N_{cell}	number of sampled cells or points on an evaluation grid satisfying a declared wall predicate	Rubik 2D slice: one Σ_{comm} hit on the chosen 20×20 evaluation grid
Repair index R_W	$D_{\text{repaired}}/\text{frozen}_{R_1}$ when both are defined	Clifford+CNOT: $6/6 = 1.0$; Pauli: $0/8 = 0$
Repair persistence p_W	parameter-range length, measured relative to the normalized deformation parameter, for which a repaired channel remains accessible after activation	CNOT-strength interpolation: threshold 0.55, $p_W = 0.45$, stability 100%
Oscillation index O_W	number of sign changes or oscillatory reversals in a plateau diagnostic	Rubik generator-weight plateau: 3; Yang-like state mixing: 0
Plateau index P_W	number, length, or collapse profile of plateau intervals in $P_d(t)$	Yang-like and training-coupled diagnostics supply initial plateau profiles
Depth index $\bar{D}$	maximum recorded D over sector pairs, with cutoff-unreached entries marked separately	some audits serialize cutoff-unreached entries as the sentinel 999; this is not proof of global impossibility; transformer diagnostic has $\bar{D} = 2$
Rate ratio ρ_W	slow observable time divided by fast observable time	NN GD+WD proxy: about $2\times$; ridge row/null diagnostic: about $68553\times$
Spectral gap γ_W	normalized second-gap or transition-gap diagnostic for a registered species	Clifford+CNOT registry example: about 0.73; Pauli control: 0
Codimension c_W	local or computational codimension when an ambient geometry is available	Σ_{comm} : 11; A_1 : 1; A_2 : 2
Wall density δ_W	fraction of species in the Paper XI wall-density taxonomy sample exhibiting a given wall type	15-entry sample shown below

These quantities make the taxonomy operational. A later SOF diagnostic report can emit a wall spectrum together with N_{cross} , N_{evt} , N_{cell} , R_W , P_W , O_W , $\bar{D}$, ρ_W , γ_W , c_W , p_W , and δ_W whenever the corresponding observable layer is defined. For example, a transformer diagnostic need not prove a wall theorem; it may report crossing count, wall-cell count, repair index, plateau or oscillation index, and depth index as observable features of the model under the chosen sectorization.

5 Wall Coordinates

Taxonomy classes record wall mechanism. They do not by themselves record where the wall lies, which observables increase or decrease, whether the realized shadow is smooth or discrete, or whether the evidence is deterministic, ensemble-defined, or pathwise stochastic. Paper XI therefore associates to a wall record the profile

$$\text{Prof}_W(\mathcal{W}) = (\text{Class}, \text{Pos}, \text{Ori}, \text{Reg}, \text{Sto}).$$

The five entries have different roles. **Class** is a mechanism label from the A–F taxonomy. The remaining four entries are realization coordinates.

5.1 Position

Pos records how the wall sits in the declared deformation domain:

Position type	Meaning
endpoint	the wall is attained only at the endpoint of the sampled path
interior	the change locus is bracketed on both sides inside the domain
boundary	the wall lies on a stopping, absorbing, or realization boundary
transition window	an ensemble or finite-resolution audit localizes an interval rather than a single point

Position is relative to the declared deformation and sampling protocol. An endpoint in one restricted path may be an interior point in a larger parameter space.

5.2 Observable Orientation

Ori records the dominant observable response. It is generally a vector rather than a sign:

$$\text{Ori}(\mathcal{W}) = (\Delta\text{gap}, \Delta R_1, \Delta R_2^{\text{word}}, \Delta R_2^{\text{Lie}}, \Delta D, \Delta P_d, \Delta \text{Term}, \dots).$$

Typical orientations include spectral closure, monotone opening or repair, freezing or contraction, plateau deformation, and terminal-component loss. This prevents a spectral gap closure from being mislabeled as negative accessibility and permits different coordinates to move in different directions across the same wall.

5.3 Regularity

Reg records the regularity of the realized wall shadow:

Regularity	Diagnostic interpretation
smooth	a smooth observable field meets a local discriminant
piecewise smooth	smooth branches meet at kinks or switching surfaces
discrete	the native deformation or observed shadow changes by jumps
degenerate	the tested trajectory is flat, unresolved, or nontransverse
algebraic/incidence	a rank, image-kernel, or bridge-product relation controls the locus

Regularity belongs to the wall record, not permanently to the source species. A smooth native flow may produce a thresholded discrete shadow, and a discrete species may admit a continuous weighting deformation.

5.4 Stochastic Status

Sto records how randomness enters the evidence:

Status	Meaning
deterministic	the declared realization and trajectory are fixed
constructed-random	a random direction or instance is sampled and then held fixed
ensemble	the wall is defined through a distribution of realizations
pathwise stochastic	individual stochastic trajectories define the native record
finite-time estimator	the wall is inferred from a sampling-dependent empirical observable

These labels are claim-status aids. They do not impose a probability geometry on the wall-coordinate space.

5.5 Class C: Terminal-Structure Walls

Let $\text{Term}(\mathcal{F}_t)$ denote the terminal decomposition induced by the declared dynamics and observable realization. Its components may be attracting basins, absorbing or recurrent classes, dead components, or stopping regions.

Definition 5.1 (Class C terminal-structure wall).

A Class C wall is a deformation locus at which $\text{Term}(\mathcal{F}_t)$ is not locally equivalent: the number, identity, incidence, or persistence of terminal components changes.

The compatible sectorization may remain fixed. Class C does not require the number of projectors or coarse sectors to change. A changing sector count is a realization change or an additional nonsmooth/discrete coordinate; it is not, by itself, the Class C mechanism. This definition includes absorbing Markov classes, barrier/stopping regions, and deterministic GRN basin loss without identifying their native dynamics.

5.6 Four Canonical Profiles

The four extended profiles give a two-axis spread in orientation and regularity, together with distinct stochastic status:

Profile	Class	Position	Orientation	Regularity	Stochasticity
constructed real- symmetric endpoint	A	endpoint discriminant	two order-one pair-gap closures	smooth conical local model	constructed-random direction
nested percolation	B/E	interior transition window	monotone opening and bounded-depth repair	pathwise discrete, ensemble- smoothed	ensemble
Kuramoto synchro- nization	D/E	finite-size interior crossover	freezing and occupancy contraction	smooth flow, thresholded shadow	matched ensemble
GRN regulatory- edge weakening	C	interior basin-loss bracket	terminal components $2 \rightarrow 1$; terminal sector loss	smooth deterministic flow, discrete terminal shadow	deterministic; stochastic noise sweep retained only as a negative control

The four profiles share one record schema while retaining distinct coordinate profiles, orientations, and regularity types.

The term **wall-coordinate space** is intentional. No equivalence relation, topology, metric, curvature, coordinate-change law, or transport structure has yet been imposed. Calling the present object a wall phase space would therefore be premature.

6 Observable Wall Taxonomy

Paper XI organizes signatures into a taxonomy before attempting any local smooth model.

6.1 Taxonomy Classes

Class	Name	Diagnostic pattern	Typical examples
A	Collision walls	spectral or discriminant branches collide or merge	Rubik spectral deformation, smooth spectral probes, NCG spectral-block diagnostics
B	Repair walls	an inaccessible or frozen channel becomes accessible at a higher observable layer	Rubik repair, quantum D -repair, transformer sparse repair
C	Terminal-structure walls	the number, identity, incidence, or persistence of attracting, absorbing, dead, recurrent, or stopping components changes	GRN basin loss, absorbing Markov classes, barrier/stopping regions
D	Plateau walls	observables remain flat, delayed, oscillatory, or degenerate over intervals	Yang-like degeneration, training plateaus, grokking-style delayed rates
E	Nonsmooth or discrete walls	wall is induced by a discrete jump, rank selection, or piecewise-smooth kink	graph rewiring, ReLU kinks, Top-k activation selection
F	Bridge or incidence walls	bridge products, rank incidence, or algebraic association controls wall behavior	Rubik Type III/IV mechanisms, incidence products, bridge-level audits

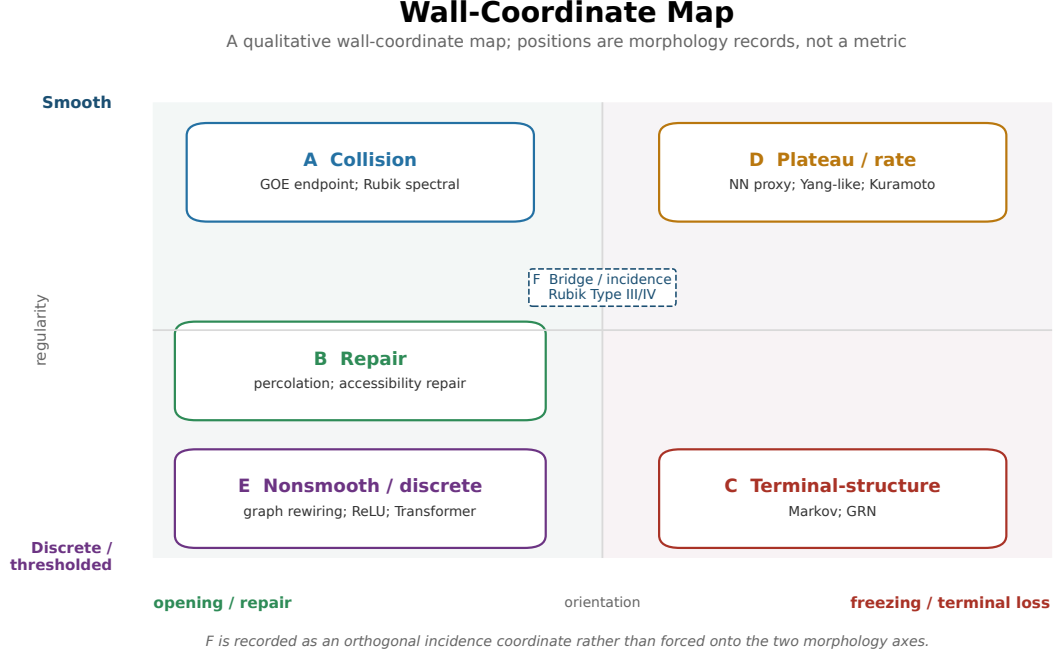


Figure 2. Wall-coordinate map. The qualitative map places registered observable morphologies by qualitative orientation and regularity. A–E are shown on the two morphology axes; F is kept as an orthogonal bridge/incidence coordinate. Positions record qualitative morphology only. The map is not a metric, a phase space, or a classification of native systems.

7 Census Evidence

7.1 Wall-Density Taxonomy Sample

In the 15-entry Paper XI wall-density taxonomy sample, wall density is computed as

$$\delta_W(\text{type}) = \frac{\#\{\text{taxonomy-sample species carrying that wall type}\}}{15}.$$

The table below is generated by the wall-density support artifact listed in Appendix C. The resulting finite-sample density table is:

Wall type	Density	Dominant source
B repair	33.3%	5 species; the most common wall type
F bridge/incidence	20.0%	Rubik naturally carries bridge and incidence walls
C terminal-structure	13.3%	barrier option and absorbing Markov examples
D plateau/rate	13.3%	Xu ridge and mechanism-separated dynamics
E nonsmooth/discrete	13.3%	graph rewiring and ReLU kink diagnostics
A spectral collision	6.7%	Rubik spectral collision branch; rarest in this sample
no registered wall type	20.0%	3 of 15 entries are static or structural species

The snapshot is not a theorem and should not be read as a population estimate for all possible SOFs. It is a taxonomy-sample statistic. Its immediate interpretation is useful: repair walls are the most common in the taxonomy sample, echoing the Paper VII focus on generic completion and repair; spectral collision walls are the rarest, suggesting that Paper XI requires more continuous deformation models before spectral-collision taxonomy can be considered representative.

Wall-Density Snapshot and Repair Persistence

First-pass Paper XI taxonomy evidence, not a population estimate or a universal quantum-wall theorem

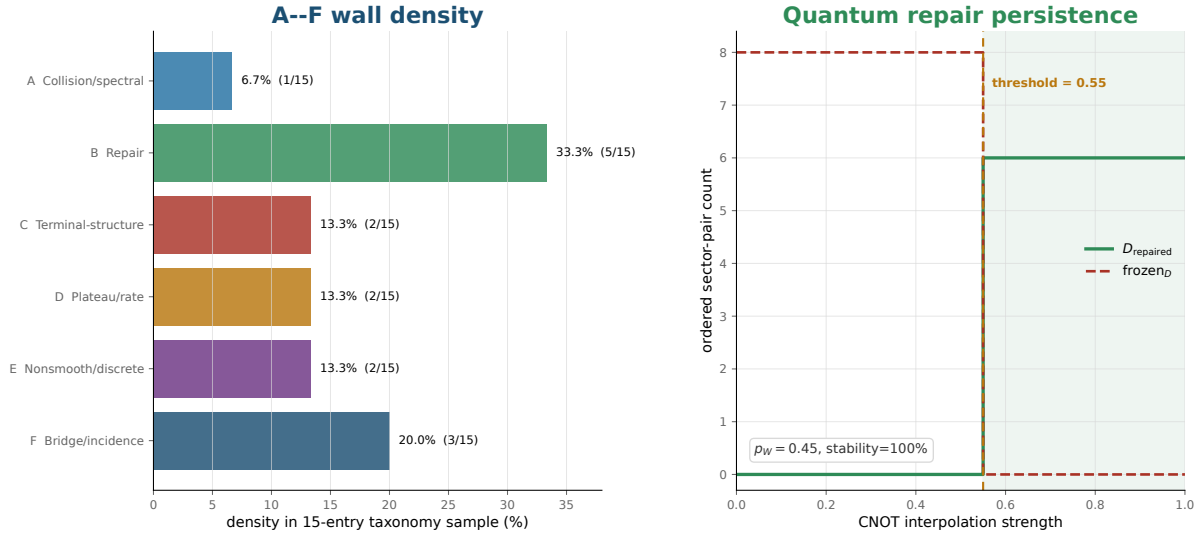


Figure 3. Wall-density and repair-persistence evidence. The left panel reports A–F wall densities in the 15-entry Paper XI taxonomy sample, which is distinct from the frozen 16-entry Paper X Registry release snapshot. The right panel shows the CNOT-strength repair threshold at 0.55, persistence $p_W = 0.45$, and 100% post-activation stability on the tested grid.

7.2 Wall-Record Coverage Census

Species prevalence alone does not measure whether each taxonomy class has enough independent wall records. We therefore maintain a second computational census whose unit is a wall record, not a species. A record may carry multiple class labels when one deformation has several observable features, but repeated records do not create new species in the prevalence denominator.

For this record-level audit, a record is **eligible** when it has an explicit deformation, an explicit change locus, a measured signature, and an existing evidence file. The predeclared coverage target for each class is at least three eligible records, two distinct species, and two deformation origins. This is a curation threshold, not a classification theorem criterion.

The original census contains 24 records. The independently generated extended census retains all 24 records and adds four controlled profiles: the constructed real-symmetric endpoint, nested percolation, Kuramoto synchronization, and GRN regulatory-edge weakening. It contains 28 records, 38 class memberships, and 19 eligible records:

Class	Registered records	Eligible records	Eligible species	Deformation origins	Coverage
A	4	3	2	3	pass
B	10	6	6	6	pass
C	5	4	4	4	pass
D	5	5	5	5	pass
E	6	5	5	5	pass
F	8	3	3	3	pass

All six classes now meet the predeclared record, species, and deformation target. This is **coverage closure**, not taxonomy completeness. In particular, Class A closes through a

constructed real-symmetric endpoint whose degeneracy is placed explicitly on the codimension-two discriminant. It is a non-Rubik species witness, but not a naturally occurring generic GOE crossing. Naturally occurring non-Rubik Class A breadth remains open. Full generated tables and record provenance are stored in the repository artifacts listed in Appendix C.

7.3 Definition-Compatible Redundancy Audit

The listed wall diagnostics do not share one sampling unit. Codimension requires an ambient geometric model, δ_W is a cross-species statistic, and O_W , P_W , and p_W require trajectories. They therefore must not be inserted into one snapshot correlation matrix by replacing them with unrelated proxies.

The definition-compatible redundancy audit instead computes eight snapshot diagnostics on the same 166 controlled configurations: direct-frozen fraction, Lie-terminal fraction, repair index, word-bridge only fraction, Lie-bridge-only fraction, mean and maximum word depth, and log observable norm ratio. The strongest empirical correlations are:

Pair	Correlation r
direct-frozen fraction / mean word depth	+0.932
mean / maximum word depth	+0.938
direct-frozen fraction / maximum word depth	+0.923
direct-frozen / Lie-terminal fraction	+0.886
Lie-terminal fraction / mean word depth	+0.856

The Lie-bridge-only fraction has zero variance on this ensemble and is removed before standardization. PCA on the seven retained diagnostics gives three components for both 90% and 95% of the observed variance. The leading axis is a reachability/depth scale; later axes separate repair, bridge-channel, and observable-rate effects. This is an empirical dimension estimate for the tested ensemble, not a proof of a minimal, complete, or orthogonal invariant basis. It is also ensemble-composition dependent: changing the relative mix of GridWorld, SIR, random-graph, and random-skew configurations can change the explained-variance percentages.

Trajectory diagnostics are reported separately. The present three trajectory controls are insufficient for a trajectory-invariant PCA, so no independence claim is made for oscillation, plateau, or persistence coordinates.

8 Observable Normal Forms and Wall Layers

8.1 SOF-Native Observable Normal Forms

Before invoking smooth singularity theory, wall signatures can be grouped into SOF-native observable normal forms. These are descriptive normal forms for observable signatures, not theorem-level ADE classes.

Normal form	Signature pattern	Typical diagnostic
NF-1: single rank collision	one spectral/rank component collides or merges	one A_1 -like gap closure or one rank jump
NF-2: bridge creation	a previously frozen pair becomes accessible at R_2 or D	positive D_{repaired} or bridge support
NF-3: oscillatory repair	repair or plateau observable alternates before stabilizing	nonzero oscillation index O_W
NF-4: plateau collapse	plateau interval degenerates, flattens, or terminates	monotone degeneration or endpoint collapse
NF-5: terminal absorption	a state, sector, or channel enters an absorbing/frozen terminal regime	frozen pairs with no repair
NF-6: bridge/incidence association	a bridge product, image-kernel relation, or algebraic incidence controls the wall	Rubik Type III/IV and rank-protected bridge audits

Observable wall records are first sorted by five axes.

8.2 Smoothness Type

Type	Description	Classification status
Smooth discriminant	$O(t)$ is smooth and Σ_O is a discriminant pullback	ADE may apply locally
Piecewise-smooth wall	$O(t)$ is continuous but not smooth across strata	needs stratified or piecewise theory
Discontinuous jump	$O(t)$ changes by a discrete event or rank jump	outside classical ADE
Degenerate/nonresponsive	trajectory is flat, decreasing, zero, or not a response curve	not a wall class by itself
Incidence mechanism	wall is defined by algebraic product or rank incidence	requires algebraic-geometric tools

8.3 Observable Layer

Layer	Typical observable	Typical wall type
Spectral	eigenvalues, collision quotients	spectral collision/discriminant
Accessibility	$R_1, R_2, D, \mathcal{J}_{\text{acc}}$	rank, support, or accessibility discriminant
Filtration	plateau functions $P_d(t)$	degeneration or plateau transition
Markov	communicating-class or transition support	absorbing/frozen boundary
Graph	Laplacian gap, adjacency transport	discrete rewiring or spectral sensitivity
Quantum	gate-log accessibility, controllability	channel/accessibility transition
Neural-network	activation sectors, $K_0/K_1/K_2$	training-coupled or activation-induced wall

8.4 Deformation Geometry

The same observable ladder can produce different wall behavior depending on what changes: generator weights, state mixing, gate interpolation, edge rewiring, training dynamics, or rate-matrix perturbation.

Therefore wall classification must be attached to a deformation geometry, not only to the SOF object.

9 Smooth Wall Models

Smooth wall models are the final layer, not the starting point. Arnold ADE classification is a plausible local model only for sufficiently smooth discriminant branches. A typical admissible situation would have:

$$O : T \rightarrow E, \quad \Sigma_O = O^{-1}(\Delta),$$

with O smooth and Δ a stable local singularity.

In this restricted setting, the local wall may fall into an A_k , D_k , or $E_{6,7,8}$ class.

This paper does not assert such a classification globally. In particular:

1. graph rewiring is discrete;
2. Top-k and step activations are discontinuous;
3. ReLU-type walls are piecewise-smooth;
4. accessibility incidence can be algebraic rather than ADE-like;
5. flat or decreasing trajectories are not rate-response walls.

ADE is therefore a local smooth-branch model, not the SOF classification framework.

The branch-aware A_n adjacency falsification audit is recorded in Appendix A. Its role in the main text is only to prevent a false promotion of sorted-index pair-gap responses into an ADE adjacency claim.

10 Cross-Species Audit

The computational audit combines cross-species wall diagnostics, smooth spectral pair-gap responses, record coverage, snapshot redundancy, branch-aware adjacency falsification, sampled trajectory events, and two registry-handoff diagnostics. The artifact map is collected in Appendix C. The discriminant-bifurcation scan is an auxiliary 2D-slice diagnostic; it is not a full map of Σ_{access} .

10.1 Rubik Spectral Snapshot

The Rubik A_{18} audit reports six spectral layers with eigenvalues

$$1, 8/9, 7/9, 2/3, 5/9, 1/3,$$

and k -set $\{0, 1, 2, 3, 4, 6\}$. This is a fixed spectral snapshot. It is not, by itself, a moving wall.

10.2 Rubik Spectral Pair-Gap Local-Model Audit

The spectral-deformation audit varies one QT generator weight and tracks the eigenvalues of a fixed separating Hermitian probe

$$M(\alpha) = QT(\alpha) + \beta HT(\alpha), \quad \beta = 0.314159.$$

The audit detects 16 pairwise adjacent-gap closures at the canonical endpoint. Each closure is recorded as an A_1 /fold candidate in the smooth spectral branch. The endpoint itself is higher

order: the regular QT/HT sector point is a simultaneous multi-branch collapse rather than a single isolated fold.

The same audit also runs a two-weight diagonal search and records simultaneous pair-gap responses. The observed count ranges from one to three across tested sampling resolutions, so the integer is not promoted to a stable invariant. The two gaps need not share an eigenvalue branch, and the audit does not perform eigenbranch continuation. These observations therefore remain pair-gap response diagnostics; they are not identified as A_2 /cusp unfoldings and do not prove an ADE classification theorem for SOF walls.

10.3 Rubik 2D Σ_{comm} Slice

The discriminant-bifurcation audit scans a two-dimensional plane of QT generator weights and tests the Frobenius norm of

$$[QT(w), HT(w)].$$

On the 20×20 slice varying two QT generator weights, the audit reports one hit on the chosen 20×20 evaluation grid. The single commutative cell has 9 spectral points, and the audit records no spectral point-count bifurcations inside the commutative cells. This is not a proof of the codimension-11 commutativity theorem from Paper VI and not a map of Σ_{access} . Its role in Paper XI is narrower: it gives a sampling anchor showing that arbitrary two-dimensional generator-weight slices meet the high-codimension commutativity locus sparsely. Spectral and accessibility wall taxonomy should therefore be computed on normal spectral charts or on deformation models chosen for that purpose, not inferred from a generic 2D scan alone.

10.4 Quantum Accessibility Contrast

The quantum audit compares Pauli, Clifford+CNOT, and Universal+CNOT gate-sector SOFs. Clifford+CNOT and Universal+CNOT show D -repair, while the Pauli control does not. This is an accessibility contrast outside Rubik, not a claim that quantum systems obey Rubik wall theory.

A separate repair-persistence audit interpolates CNOT strength in matrix space from a product endpoint to the Clifford+CNOT gate-log SOF. On the 21-point grid $0, 0.05, \dots, 1$, D_{repaired} is zero through strength 0.50 and jumps to 6 at strength 0.55. The repair then persists through strength 1.00 with no reversal. Thus the registered repair-persistence signature is threshold 0.55, $p_W = 0.45$ over the unit interpolation range, and post-activation stability 100%. In grid-count terms this corresponds to 10 active samples out of 21; p_W records the continuous parameter range, not the sample fraction.

10.5 Markov Frozen-Pair Contrast

The Markov audit separates complete/strongly connected examples from an absorbing-state example. The absorbing case has true frozen pairs. This is a communicating-class or absorbing-boundary diagnostic, not an ADE classification.

10.6 Graph Spectral-Gap Sensitivity

The graph audit records Laplacian-gap changes under edge removal. This is a discrete perturbation diagnostic. It is not a smooth codimension-one wall without a separate continuous edge-weight model and discriminant calculation.

10.7 Piecewise-Smooth Activation Wall Boundary

The activation-wall audit compares ReLU, GeLU, and Top-k sector diagnostics. In the default finite test, the ReLU sector diagnostic has distinct left and right slopes, producing a piecewise-smooth kink. GeLU is smooth in the same test and does not define a kink wall. Top-k is a rank-selection diagnostic and is treated as a discontinuous or discrete wall record.

This audit supports the negative boundary for ADE usage: activation-induced SOF walls may require piecewise-smooth, stratified, or discrete wall theory rather than classical smooth ADE singularity theory.

10.8 Barrier-Option Stopping Boundary

The barrier-option SOF uses a finite log-price grid with below-barrier and above-barrier sectors. The default audit reports cross-barrier generator support with $R_1 = 75.0\%$, $R_2 = 0.0\%$, no D -repair, and mean first-hit time 6.5915 from the chosen initial grid point to the barrier sector. This is a stochastic stopping-region wall record. The first-hitting time is a native stochastic diagnostic, not the SOF depth D , and the entry is not an option-pricing theorem.

10.9 Transformer Activation-Sector Diagnostic

The transformer diagnostic constructs a token-space SOF from activation-count clusters in a small synthetic transformer-like block. Attention top-k and FFN activation-similarity operators give $R_1 = 58.3\%$, $R_2 = 66.7\%$, $D_{\text{repaired}} = 2$, and $D_{\text{max}} = 2$ in the default audit. This is an activation-induced diagnostic wall record: it shows how token-sector decompositions can expose cross-sector influence, but it is not a theorem about all transformers or LLM explainability.

10.10 Constructed Real-Symmetric Endpoint

Setup. The non-Rubik Class A witness is the family

$$H(t) = D + tV, \quad D = \text{diag}(1, 1, 2, 3, 4, 4),$$

with V a sampled GOE splitting direction. The endpoint D is deliberately placed on the real-symmetric double-eigenvalue discriminant.

Observed wall record. The two isolated target gaps have fitted local orders 1.000015 and 0.999968 in the canonical draw, with first-order splitting coefficients 0.598557 and 0.333281. The minimum non-target gap on the tested local interval is 0.937503. Four sampled transverse splitting directions pass the local-order and endpoint-lifting controls.

For a 2×2 degenerate block, the traceless normal coordinates (x, y) give

$$\text{gap} = 2\sqrt{x^2 + y^2}.$$

The one-parameter family follows a ray through this conical normal slice, so the pair gap opens linearly. Endpoint-preserving changes of the splitting direction remain transverse in the tested draws, while perturbing D off the discriminant removes both collisions.

SOF interpretation. This is a constructed Class A A_1 -type endpoint witness: a smooth local collision record with two order-one pair-gap closures.

Boundary. This is not a generic one-parameter GOE crossing and not an unrestricted structural-stability theorem. It closes the predeclared finite-sample Class A coverage target through a constructed witness; naturally occurring non-Rubik Class A breadth remains open.

10.11 Nested Percolation Opening

Setup. For each Erdos–Renyi ensemble member, one symmetric threshold matrix $U = (U_{ij})$ is fixed and

$$G(p) = \{ij : U_{ij} < p\}.$$

The path is nested: edges are added but never removed. Raw adjacency is used as the nonnegative word-transport observable.

Observed wall record. In the default 32-member ensemble, the largest mean bounded-depth frozen-count drop occurs at $p = 0.08$, while the largest ensemble fluctuation occurs at $p = 0.10$. Across the robustness audit, the opening window lies in $p \in [0.08, 0.10]$. Direct support and bounded-depth reachability are verified to be monotone for every nested realization.

SOF interpretation. This is a B/E profile: a monotone repair wall in orientation, with a pathwise discrete record and an ensemble-smoothed transition window.

Boundary. The result is pathwise discrete and ensemble-smoothed only. Degree normalization followed by skew-symmetrization would permit cancellation and would destroy the monotone support control. This is not a theorem about arbitrary percolation processes or a claim that all opening walls have the same local geometry.

10.12 Kuramoto Freezing Orientation

Setup. The Kuramoto control fixes each natural-frequency sample and initial phase vector across the complete coupling sweep. Frequencies are centered so that the (r, ψ) sectorization is evaluated in a common co-rotating frame. The raw directed trajectory transition is the word observable.

Observed wall record. In the default eight-member matched ensemble, the largest mean freezing step occurs at $K = 1.6$. Between $K = 0.8$ and $K = 2.4$, the bounded-depth frozen count increases by 273.4 ± 38.0 , with positive direction in every tested member. Across the robustness audit the crossover lies in $K \in [1.6, 1.8]$. The continuum Gaussian synchronization scale $K_c \approx 1.277$ is only a reference.

SOF interpretation. This is a D/E profile and a negative-orientation wall: stronger coupling produces occupancy contraction and more frozen sector pairs. Wall orientation therefore cannot be identified universally with repair or increasing accessibility.

Boundary. The SOF wall location depends on finite size, time window, sector resolution, and depth cutoff. This is not a synchronization theorem, nor a universal statement about Kuramoto criticality.

10.13 GRN Terminal-Structure Loss

Setup. The GRN positive record varies the strength λ of one repression edge $A \dashv B$ continuously from 1 to 0 in a deterministic toggle-switch flow. The compatible 8×8 concentration sectorization is fixed.

Observed wall record. At $\lambda = 1$ the controlled initial grid reaches two attracting basins; after the edge is removed only one remains. The coarse scan brackets the basin loss in $[0.50, 0.55]$, and the controlled-initial refinement gives $[0.520, 0.525]$. The terminal-sector set changes from $\{5, 40\}$ to $\{5\}$.

Methodological correction. The initial-release low- Ω noise-wall interpretation is withdrawn. Under a matched fixed-time protocol with the same concentration grid and raw directed observable, a 3-seed by 12-trajectory comparison gives CLE and exact SSA frozen-depth ranges of 0.94% and 0.89%, respectively, a maximum method gap of 6.7/4032, and no SSA basin switches for $\Omega = 60, 30, 15, 10$.

SOF interpretation. This is the canonical Class C terminal-structure wall: terminal components change from 2 to 1 and one terminal sector is lost. The positive evidence is deterministic terminal-structure loss, not stochastic basin merging.

Boundary. The refined interval is a numerical diagnostic, not a saddle-node theorem. The fully deleted edge is a discrete endpoint intervention, while the primary wall is the interior change in the terminal decomposition along the continuous λ deformation. No stochastic basin-merging or absorbing-state theorem is claimed. The superseded discrepancy resulted from mixing event-count sampling with incompatible sector and observable choices.

10.14 Extended Profile Robustness

The unified extended-profile robustness audit evaluates the same canonical implementations used for the individual profiles. Four sampled GOE splitting directions pass the local-order and endpoint-lifting controls; nested percolation gives a wall window $[0.08, 0.10]$; matched Kuramoto controls give a freezing window $[1.6, 1.8]$; and the GRN terminal-component loss remains in the controlled-initial bracket $[0.520, 0.525]$. These are protocol-scoped robustness statements, not universality results.

11 Supporting Audit Boundaries

Two boundary audits are deferred to the appendices. Appendix A records the branch-aware adjacency falsification audit for Rubik spectral pair responses. Appendix B records the observable-status trajectory audit. Their shared main-text role is cautionary: sorted gaps should not be promoted to ADE adjacency without branch continuation, and static inaccessible pairs should not be relabeled as walls without a change along a declared deformation path.

12 Taxonomy Registry Summary

This compact registry view records only species or realization, assigned wall class, and claim boundary. It does not repeat the experimental descriptions or replace the record-level claim-status data in the five-layer census.

Species / realization	Wall class	Claim boundary
Rubik spectral snapshot	A reference: fixed finite arrangement	Exact snapshot; not a moving wall by itself.
Rubik spectral deformation	A: higher-order endpoint with 16 pairwise A_1 closures	Smooth-branch diagnostic; no tested $A_2 \rightarrow A_1 + A_1$ split.
Rubik Σ_{comm} slice	A/B anchor: sparse sampled intersection	Boundary diagnostic; codimension 11 belongs to Paper VI.
Accessibility $R_1/R_2/D$	B: support or first-depth repair jump	Conditional on a declared deformation and threshold.
Rubik Type III/IV mechanisms	F: cancellation and rank-incidence structure	Algebraic evidence; not ADE by default.
Quantum accessibility	B: entangling-gate D -repair	Diagnostic evidence; local model unclassified.
Markov absorbing boundary	C: unrepaired terminal component	Diagnostic evidence; not ADE by default.
Graph edge rewiring	E: discrete gap and support change	Outside smooth ADE.
Barrier-option stopping sector	C/D: first-hitting boundary	Registry diagnostic; hitting time is not SOF depth.
NN training dynamics	D: delayed or ordered proxy response	Paper IX/X proxy evidence; rate order is not a wall class.
ReLU / Top-k activations	E: kink or rank-selection jump	Requires nonsmooth or stratified models.
Transformer activation sectors	B/E: sparse token-sector repair	Diagnostic handoff; activation-derived sectorization.
Yang-like state mixing	D: plateau degeneration	Comparison branch; model-dependent smoothness.

13 Scope and Claim Boundary

Paper XI classifies **observable wall records**. It does not classify the underlying dynamical systems, phase transitions, singularities, PDE bifurcations, or catastrophe-theory objects themselves. These native structures may be related to a wall record, but they are not identified with it.

Paper XI intentionally avoids introducing a formal wall geometry. The wall-coordinate map is a qualitative visualization of observable morphology, not a metric, topology, curvature theory, or coordinate-change structure.

Different systems may generate identical wall records, while the same system may generate different wall records under different observable families or sectorizations. The record language is therefore intended for comparison of observable morphology, not for replacing the native theory of a registered species.

Accordingly, Paper XI does not claim:

1. a complete classification of all SOF trajectories or their native systems;
2. that Arnold ADE applies to all SOF branches, including graph rewiring or activation-induced walls without extension;
3. that the mechanism-separation principle or rate hierarchy alone determines wall class;
4. that Paper XI supplies general transformation laws for observables across walls;
5. that the wall assignment is full, faithful, or invariant under all weak SOF comparisons;

6. that the listed diagnostics form a complete invariant system or that the observable normal forms are classification theorems;
7. that the tested Rubik slices exhibit A_n adjacency or that the constructed endpoint is a generic GOE crossing;
8. that finite-sample census coverage is taxonomy completeness or a population estimate;
9. that the wall-coordinate space already has a metric, topology, curvature, coordinate-change law, or transport structure;
10. that the GRN audit establishes a stochastic basin-merging or absorbing-state theorem.

The stable claim is narrower:

SOF wall behavior is first organized by observable wall records, wall signatures, wall spectra, computable observable diagnostics, and wall-coordinate profiles. The Observable Wall Taxonomy groups those signatures into collision, repair, terminal-structure, plateau, nonsmooth/discrete, and bridge/incidence classes. The extended census closes the predeclared record, species, and deformation target for every class, while retaining explicit constructed-evidence and natural-occurrence boundaries. Observable normal forms organize the registered signatures before any smooth singularity model is invoked. ADE-type local modeling is one candidate subtheory for sufficiently smooth wall records, not the whole theory.

14 Conclusion

Paper XI is about **observable wall morphology**, not universal wall dynamics. Its purpose is to organize observable wall morphology rather than derive universal wall-transition laws. It turns SOF wall phenomena into a proto-geometric record-level classification problem. Its basic output is not a universal wall-crossing law, but a sequence of observable records, signatures, wall spectra, wall diagnostics, and wall-coordinate profiles that can be compared across registered SOF species. Mechanism classes answer what kind of wall occurs; position, orientation, regularity, and stochasticity answer how that mechanism is realized. Smooth ADE-style normal forms remain candidate local models only on sufficiently smooth discriminant branches; the taxonomy itself also admits repair, terminal-structure, plateau/rate, nonsmooth/discrete, and bridge/incidence records.

The four extended profiles show why both layers are needed. Spectral gaps may close, accessibility may open, synchronized dynamics may freeze, and terminal components may disappear. The record language is common; the realized geometry is not. The extended census reaches the predeclared finite-sample coverage target for A–F, but coverage closure is not classification completeness.

15 Outlook

Paper XI supplies a wall-coordinate taxonomy, not a classification theorem. The next stage is to determine which coordinates are stable under comparison of SOF presentations, which depend on the chosen ladder or sampling path, and which admit local geometric normal forms.

The remaining classification program has seven parts:

1. develop or import piecewise-smooth and stratified wall theory for activation-induced and rank-selection SOFs;
2. formalize comparison and equivariance conditions for $\text{WallAssign}(\mathcal{F}, \mathcal{L}, \Gamma)$: defining compatible comparison data is straightforward, but proving invariance under weak SOF comparisons, path reparametrizations, and sampling refinements is a separate theorem-level problem;
3. define equivalence and admissible coordinate changes for wall profiles;
4. test normalization and sampling stability before introducing any wall metric;
5. build continuous graph and Markov deformation models where smooth discriminants can be computed;
6. compute local codimensions for accessibility discriminants and higher-order spectral endpoints;
7. separate incidence varieties from smooth singularity classes.

Only after these ingredients and additional registered examples are available can the taxonomy be promoted toward classification theorems.

Appendix A Branch-Aware Adjacency Falsification Audit

Simultaneous small gaps in a sorted full-spectrum list are not enough to establish A_n adjacency. Such gaps may belong to different invariant blocks, may exchange order, or may be symmetry-protected degeneracies rather than isolated collision branches.

The branch-aware audit therefore separates the Rubik representation into its cp , ep , co , and eo blocks and continues eigenbranches by maximizing eigenvector overlap between adjacent parameter samples. A candidate $A_2 \rightarrow A_1 + A_1$ split requires two isolated pair-gap minima at separated parameter values whose union involves all three branches. Persistent degeneracies, endpoint-only closures, and cross-block equalities are rejected.

Block	Dimension	Endpoint cluster sizes	Tested triples	Split candidates
cp	64	8, 24, 24, 8	0	0
ep	144	12, 36, 24, 36, 36	0	0
co	8	3, 3	2	0
eo	12	3, 3, 3	0	0

On the tested diagonal and asymmetric slices, no branch-aware $A_2 \rightarrow A_1 + A_1$ split candidate is detected. The sorted-index pair-gap responses remain useful spectral diagnostics, but they do not support an A_n adjacency claim. Even a future positive candidate would still require local normal-form and versality analysis before becoming an ADE statement.

Appendix B Observable-Status Wall Trajectories

Given audit snapshots at parameters $t_0, \dots, t_K$, assign each ordered pair one observable status:

$$S_{ij}(t) \in \{\text{direct, word bridge, Lie bridge, deeper, terminal}\}.$$

A sampled wall event occurs only when

$$S_{ij}(t_{k-1}) \neq S_{ij}(t_k).$$

The trajectory implementation records every change between adjacent samples, including repeated changes of the same pair. Events are labeled as repair, terminalization, support gain, support loss, or layer change. Stable terminal pairs remain trajectory metadata and are not counted as walls.

Three controlled paths are implemented:

Control	Steps	Ordered pairs	Events	Changed pairs	Event steps
GridWorld obstacle path	3	600	190	140	1, 2
SIR β sweep	21	6	4	4	1
Graph edge-weight endpoint	11	12	6	6	10

The SIR result isolates all changes at the first sampled step $\beta = 0 \rightarrow 0.025$. The graph control is constant for $t < 1$ and changes only when the weakened edges vanish at $t = 1$. GridWorld records two discrete topology replacements and therefore has events at both steps.

These are wall records on sampled paths. They do not establish ambient codimension, continuous-time wall flow, or a bifurcation theorem. Their role is narrower: they provide a correct dynamic record layer between Paper IX deformations and the Paper XI taxonomy without relabeling every frozen pair as a wall.

Appendix C Computational Artifacts

The following repository artifacts provide the reproducible support layer for the audits reported in this paper. The main text refers to audit roles rather than file paths; this appendix records the exact implementation and generated tables. Unless a prefix is shown, files are located under `experiments/paper11/`. The prefixes `validation/` and `results/` are relative to that directory; `paper10/` and `paper12/` are relative to `experiments/`.

Artifact	Role in Paper XI	File
C1	cross-species wall diagnostics	cross_species_wall_audit.py
C2	smooth spectral pair-gap local-model audit	spectral_ade_collision.py
C3	auxiliary 2D commutativity-discriminant slice	discriminant_bifurcation_map.py
C4	wall-density taxonomy sample	wall_density_registry.py
C5	original 24-record wall-record coverage census	wall_record_census.py
C6	original generated census tables	results/wall_record_census.md, .json
C7	definition-compatible redundancy audit	invariant_redundancy.py
C8	branch-aware adjacency falsification	an_adjacency.py
C9	observable-status trajectory records	wall_trajectory.py
C10	CNOT-strength repair persistence	repair_persistence_quantum.py
C11	piecewise-smooth activation boundary	piecewise_smooth_activation_wall.py
C12	barrier-option registry handoff	paper10/barrier_option_sof.py
C13	transformer activation-sector handoff	paper12/transformer_activation_sof.py
C14	constructed real-symmetric endpoint collision witness	validation/degenerate_endpoint_collision.py
C15	nested percolation opening wall	validation/percolation_wall.py
C16	matched Kuramoto freezing control	validation/kuramoto_wall.py
C17	GRN CLE/SSA negative control and terminal-basin loss	validation/grn_toggle_wall.py
C18	unified extended-profile robustness audit	validation/wall_robustness_audit.py
C19	independent extended wall-record coverage census	validation/wall_record_census_v2.py
C20	generated extended census tables	results/wall_record_census_v2.md, .json

The canonical manuscript build entry, run from the repository root, is `python papers/tex/build.py paper11`. The build consumes the frozen PDF figure assets under `figures/paper11/` and writes the compiled artifact to `papers/tex/paper11_arxiv.pdf`.

Bibliographic Notes

Program lineage. Paper XI depends on Papers VII–X. Paper VII supplies the generic completion and repair boundary [3]; Paper VIII supplies the SOF object layer [5]; Paper IX supplies observable trajectories and wall pullbacks [4]; Paper X supplies the Universal Observable Pipeline and five-layer SOF Registry [6].

Smooth singularities. Arnold’s catastrophe and ADE singularity theory is relevant only to the smooth-discriminant branch. Useful background includes Arnold’s *Catastrophe Theory* [1], the work of Arnold, Gusein-Zade, and Varchenko in *Singularities of Differentiable Maps* [2], and Golubitsky and Guillemin’s *Stable Mappings and Their Singularities* [12].

Stratified and nonsmooth walls. Piecewise smooth, discontinuous, and multi-stratum SOF walls require different tools. Relevant background includes stratified Morse theory [13], piecewise-smooth dynamical systems [8], and differential equations with discontinuous right-hand sides [10].

Spectral and finite-state diagnostics. Graph and Markov wall records are compared against spectral perturbation theory [14], spectral graph theory [7], and Markov chain perturbation and mixing theory [16] before any singularity classification is claimed.

Additional-profile precedents. The constructed endpoint audit uses standard degenerate perturbation theory and the von Neumann–Wigner codimension boundary for real-symmetric level crossings [17]. The percolation profile is interpreted against Erdos–Renyi random-graph connectivity [9], the synchronization profile against the finite Kuramoto model [15], and the

GRN method control against Gillespie’s exact stochastic simulation algorithm [11]. These external theories motivate the native deformation diagnostics; they do not supply SOF taxonomy theorems automatically.

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